Ref. Ares(2023)2336631 - 31/03/2023

Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

Project funded by the European Innovation Council and SMEs Executive Agency

D9.1 AI - Requirement No. 1

Work Package: WP9

Authors: Ana Tanesvka, Arabinda Ghosh

Project Title: Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

Project Acronym: SymAware

Grant Agreement number: 101070802

Project Start Date: 1st October 2022

Duration: 36 months

Project Coordinator: Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Project Members:

Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Technische Universiteit Eindhoven (TUE)

Kungliga Tekniska Hoegskolan (KTH)

Uppsala Universitet (UU)

Stichting Koninklijk Nederlands Lucht - En Ruimtevaartcentrum (NLR)

Siemens Industry Software Netherlands BV (SISW)

Document Identifier: D9.1

Issue: 1.2

Issue Date: 30.03.2023

Prepared by: UU

Approved: Yes

Dissemination Status: Confidential, only for members of the consortium

History Chart:

Issue	Date (DD.MM.YYYY)	Updated Sections	Cause of Change	Implemented by
1.0	29.03.2023	All sections	New document Added content	Ana Tanevska
1.1	30.03.2023	All sections	Edited content	Arabinda Ghosh
1.2	30.03.2023	Section 2	Edited content	Ana Tanevska

All rights reserved

The document is proprietary of the SymAware consortium members listed on the front page of this document. The document is supplied on the express understanding that it is to be treated as confidential and may not be used or disclosed to others in whole or in part for any purpose except as expressly authorized in terms of Grant Agreement number 101070802. The SymAware consortium makes no warranty for the information contained in this document; neither does it assume any legal liability or responsibility for the accuracy, completeness or usefulness of this information. Company or product names mentioned in this document may be trademarks or registered trademarks of their respective companies.

Contents

1. Introduction....4
1.1 SymAware: High-level description of the project....4
2. Ethical Requirements for a Trustworthy Human-Robot and Human-Agent Interaction in Social Contexts with Autonomous Vehicles and Drones....5
2.1 Research Overview....5
2.2 Research Questions and the Corresponding SymAware Tasks....6
2.3 D9.1 - Ethical Requirements....7
2.3.1 Research Procedures, Data Collection, Data Preservation and Purpose of Collected Data....7
2.3.2 Participant Handling – Informed Consent, Risks....9
Possible risks for the participants....9
2.3.3 Bias, System Auditability, and User Agency....12
3. Related SymAware Deliverables in WP9....14
3.1 Deliverable 9.2....14
3.2 Deliverable 9.3....14
Bibliography....15

1. Introduction

1.1 SymAware: High-level description of the project

SymAware addresses the fundamental need for a new conceptual framework for awareness in multi- agent systems (MASs) that is compatible with the internal models and specifications of robotic agents and that enables safe simultaneous operation of collaborating autonomous agents and humans.

The goal of SymAware is to provide a comprehensive framework for situational awareness to support sustainable autonomy via agents that actively perceive risks and collaborate with other robots and humans to improve their awareness and understanding while fulfilling complex and dynamically changing tasks.

The SymAware framework will use compositional logic, symbolic computations, formal reasoning, and uncertainty quantification to characterize and support situational awareness of MAS in its various dimensions, sustaining awareness by learning in social contexts, quantifying risks based on limited knowledge, and formulating risk-aware negotiation of task distributions. These objectives will be achieved in SymAware through:

1. logical characterization of awareness using symbolic methods,
2. quantifying the symbolic reasoning for awareness with spatial and temporal ingredients for decision making,
3. risk awareness via quantified knowledge,
4. quantifying and communicating knowledge awareness,
5. demonstrating awareness engineering in aviation and automotive use cases, and
6. identifying requirements for ethical and trustworthy awareness in human-agent interaction.

The objectives of SymAware address the “Awareness Inside” Challenge of the European Innovation Council (EIC) by extending and formalizing human-based models of situational awareness and by providing a novel conceptual situational awareness framework for MASs that encompasses logical characterization and integrative formal reasoning of interdependent awareness dimensions including knowledge, spatiotemporal, risk and social dimensions. This will support transitioning to safe mixed operation of autonomous agents and humans.

2. Ethical Requirements for a Trustworthy Human-Robot and Human-Agent Interaction in Social Contexts with Autonomous Vehicles and Drones

2.1 Research Overview

An important factor in trustworthy human-agent and human-robot interaction is the person's attribution of a mind to the artificial agent or robot [1]. However, when the robot's perceived mind and actual capabilities are unaligned, this can be detrimental to a successful interaction with the person [2, 3]. Thus, it becomes very important for the robot or agent to be transparent about its abilities, relying on explainability mechanisms which would endow it with the ability to provide information about its inner workings using social cues, in a way that the human observer/interaction partner can infer how and why the agent or robot behaves the way it does [4].

In the SymAware project, Uppsala University as a partner will explore new machine learning methods for the robot, in a way which would enable the human users to teach the robot explainable behaviours by providing demonstrations [5].

We will also investigate different classes of system-identified risks and safety measures in human-agent and human-robot interaction (HAI/HRI) [6,7], develop explainability mechanisms to communicate risk and spatial information to human users, as well as design mechanisms for promoting trust and fairness and preventing bias in the developed computational methods [8].

Finally, we will develop user-centred and participatory design (PD) methods [9] and undertake user studies to investigate the ethical aspects of artificial agents collaborating with and supporting users, human-in-the-loop monitoring of the agents' interactions, and investigate adverse effects of agents' autonomy [10].

To ground the research in concrete scenarios, we will conduct the investigations above in the areas of simulation of autonomous vehicles (AV) and traffic management of multi-drone systems. The two proposed showcases will each include instances of interaction between a human and an artificial agent/robot, both as direct interaction between the person and the AV simulator/robot/traffic management system, and indirectly during the participatory design and focus group sessions.

We plan to conduct experiments to address these main research questions:

1. *How can we identify any potential ethical issues in human-agent interaction and develop mechanisms to promote fairness?*
2. *How can we transfer our knowledge of human goal understanding to the task of explaining an agent's goals to humans?*
3. *What are the mechanisms behind indication, communication, and explanation of system-identified risk (within the context of machine operations) to another agent in an agent-human interaction?*
4. *How can we understand and implement the process of gathering and communicating spatial information to humans?*

2.2 Research Questions and the Corresponding SymAware Tasks

The abovementioned four research questions are translated to the UU research objectives as well as corresponding SymAware tasks. We outline below the methodology (and specific studies) that we will use to address them:

1. Identifying and addressing ethical issues in human-agent interaction and mechanisms to promote fairness (T6.1+6.3, M1-36).

This objective will be investigated by:

- Conducting user studies with end users in specific human-robot and human-agent interaction use cases to understand the envisioned ethical issues in the interaction;

- Investigating the concept of human oversight with user-centred design methods, researching to what extent the agent should act autonomously, how people can monitor the interaction, and how the agent could support the users without replacing them;

- Developing methods for situational awareness, following the requirement of fairness and account for inclusion and diversity, in order to avoid biases in the system;

- Investigating how to avoid gender biases and promote fairness, for example by collecting data from a balanced sample of male and female participants, at different stages of the design, development and evaluation of situational awareness technologies.

2. Human goal understanding and explaining agent's goals to humans (T2.2, M7-36).

This objective will be addressed by:

- Developing new methods for human-agent mutual understanding, in order to enable an artificial agent to understand the intention of an interacting user and the environment;

- Building a user model to identify the relevant user goals to specific scenarios, and conducting focus groups with users to investigate what mechanisms to develop to explain agent goals to users;

- Exploring whether an "agent learning from human demonstration" approach can be used for agents to learn from users about being transparent and explainable;

- Evaluating developed methods in human-agent interaction studies to test their effectiveness on user experience and interaction quality.

3. Indication, communication, and explanation of system-identified risk to another agent in the agent-human interaction (T3.2, M13-32).

This objective's methodology will consist of:

- Using PD methods including interviews, focus groups and sketching, to develop prototypes of the explainability mechanisms to effectively communicate system-identified risk in HAI;

- Evaluating HAI studies with participants in order to measure the system's perceived trustworthiness as well as user experience and quality of interaction.

4. Gathering and communicating spatial information to humans (T4.3, M21-36).

Tackling this objective will include:

- Using human-centred design methods to involve final users in the design and evaluation of methods to provide spatial information to humans.

- Conducting HAI/HRI studies with users to evaluate the effectiveness of the developed methods in terms of user experience, quality of the interaction, and user cognitive load.

2.3 D9.1 - Ethical Requirements

As Uppsala University will be the coordinating partner for all planned Human-Robot and Human-Agent Interaction studies, as per Swedish regulations, we have submitted an Ethical Risk application to the Swedish Ethical Review Authority (EPM – Etikprövningsmyndigheten). With that application as a reference point, we outline below the ethical requirements as indicated in WP9, and our methodologies for addressing and mitigating any potential ethical risks.

2.3.1 Research Procedures, Data Collection, Data Preservation and Purpose of Collected Data

Research methodology

Throughout WP 6, UU will conduct user studies, such as interactions with artificial agents or robots, interviews, and focus groups, to understand what are the envisioned ethical issues in the human-agent interaction. We will be conducting two different types of studies - participatory design (PD) studies with human participants, and human-robot/human-agent interaction (HRI/HAI) studies.

The PD methodology will follow an iterative design approach, where human users are involved in all steps of the methods' development and evaluation, and which combines different human-agent interaction methods. Participatory design methods including interviews, focus groups and sketching will be used to develop prototypes of the SymAware technologies.

Moreover, human-agent and human-robot interaction studies with human participants will be conducted in order to test the effectiveness of the developed technologies in terms of user experience and quality of the interaction. The HAI/HRI studies will be designed in the context of the use cases of the SymAware project, which is centered around simulation of autonomous vehicles (AV) and multi-drone systems. As such, some examples of the studies would involve the human participants watching and evaluating videos of a simulator of an autonomous vehicle, interacting with the AV simulator where they would be in the role of the driver, and evaluating their safety and trust in AV.

We will investigate to what extent users think that the agent should act autonomously, how can human users be in the loop and monitor the interaction, and how can the agent support users, without at the same time replacing them. To ensure transparency, prior to their participation in the studies, participants will be debriefed on what agents can do and what they cannot do (i.e. which capabilities they are endowed with and which are not).

Studies with human participants will be conducted in a laboratory environment. During the studies, no physical intervention will take place. We intend to use **non-intrusive** methods of collecting data during the interaction, such as video and audio recordings (e.g., via video cameras), eye tracking data, GSR data, and task data. Video recordings and eye tracking data will be used for visual analysis and annotation to explore how participants interact with robots.

We will also ask the participants to fill in some questionnaires to rate their overall experience with the robot and perception of it. We will also collect qualitative data regarding the participants' experiences, e.g. utilising the "think aloud" method to capture real-time participant reactions to interaction events, as well as free-text response options on questionnaires. Participants will generally interact with the robot or agent for a duration of 15-30 minutes.

Data collection and handling

We comply with the EU's data protection regulation (General Data Protection Regulation) and handle data according to national Swedish law (The Archives Act). All data is stored without information that can identify participants. Participants will be assigned a random number, i.e. a unique code (P-ID) (eg in notes or file names we refer to a participant by a number, we record the number on the consent form so we can find relevant information if necessary). The code key is saved electronically on a locked computer at Uppsala University.

The code key and all data collected from the studies are treated confidentially and stored anonymously and securely in a password-protected computer at Uppsala University and are available on request to researchers in the SymAware project, who follow the same basic protocol for data storage. The computers will be accessible from Uppsala University's network.

Uppsala University is responsible for storing and managing data. Participants taking part in the study may be videotaped. We may display video recordings of these studies in scientific articles, conferences and events, but participants can opt out by ticking the box on the consent form (see attachment). Data collected during this study, e.g. data automatically extracted from the video recordings (e.g. the position of the main points of the face, such as the corners of the mouth or eyebrows) can be shared with peers and collaborators outside the project and displayed in publications anonymized and in numerical format (i.e. no video with identifiable information will shared but only numbers representing facial features and expressions) to facilitate transparency and openness in science. Collected data will be processed for as long as necessary to ensure the quality of the research. Data is stored for the entire duration of the project and thereafter as long as required.

2.3.2 Participant Handling – Informed Consent, Risks

Possible risks for the participants

In this project we will explore the use of social robots and simulated artificial agents for developing mechanisms for ethical and trustworthy human-robot and human-agent interaction. The results of the project will inform the design of robots that interact and cooperate with humans in a more effective and natural way, particularly as it relates to the areas of risk awareness and communication, trust, learning, human agency and oversight.

The planned studies are non-invasive and, as far as we can tell, do not entail any additional risks beyond the usual ones. It is possible that participants interacting with artificial agents or robots can **assign them skills that they do not possess**. It is also possible that participants believe that the **agents' or robots' skills are limited.**

To address this, we plan an information session with our participants at the end of the study to explain how the robot is programmed, when and how it is operated by a researcher, and what limitations it possesses. The participants will be debriefed in full about their interaction with the artificial agent or robot, they will be told about the abilities, limitations, risks, and benefits of the agent/robot, and they will be educated on the decision-making process of the agent/robot as well as the underlying logic.

Before each planned study, UU will also have meetings with the Independent Ethical Advisor with whom we will discuss all of these points, and they will perform an external audit of the methodology and outline any potential risks that we may have overlooked. These reports from the advisor will be included in SymAware's periodical reports.

Participant recruitment and information process

In the beginning we mainly intend to recruit participants from Uppsala University. These may include students, researchers, and other university staff. This will later be expanded to include also more general target population. We may use posters, e-mails, and brochures to contact potential volunteers. Consent will be sought in oral and written form.

Approximately 30-40 participants will take part in each human-robot interaction study. We will recruit healthy adult (>= 18 years) participants, including both male and female participants. Participation will be voluntary and this will be clarified in the participant information sheet.

Participants will receive their own research subject information and consent form. Before the experiment begins, participants will have the opportunity to read the information sheet and ask questions of the researchers in charge. Participants will be informed that they can opt out of the study at any time. Consent will be sought in oral and written form.

Below follow example templates of the "Participant Recruitment", "Research Subject Information" and "Informed Consent" forms.

Participant Recruitment form

We are looking for people who would like to participate in a short experiment about artificial agents and social robots. The project investigates the use of artificial agents and social robots to research the concepts of ethical and trustworthy human-agent and human-robot interaction, specifically in a context where people interact with autonomous vehicles and multi-drone systems.

The study helps us understand which elements of the agent's/robot's behavior impact the people's trust and sense of agency and control in the interaction, as well as any potential ethical issues that could come up in an interaction of that kind. A session takes approximately 30 minutes.

Contact:

"*responsible researcher's contact information"*

Research Subject Information form

We wish to ask you if you want to participate in the research project *"Ethical and trustworthy interaction between human-robot and human-agent in social contexts with autonomous vehicles and drones"*. In this document you will receive information about the project and what it means to participate.

What kind of project is it and why do you want me to participate?

We are seeking adult participants to participate in this study.

The project investigates the use of artificial agents and social robots to research the concepts of ethical and trustworthy human-agent and human-robot interaction, specifically in a context where people interact with autonomous vehicles and multi-drone systems.

The study helps us understand which elements of the agent's/robot's behaviour impact the people's trust and sense of agency and control in the interaction, as well as any potential ethical issues that could come up in an interaction of that kind.

What is an artificial agent?

An artificial agent is anything (robot, machine, virtual avatar, software program) which perceives its environment, takes actions autonomously in order to achieve goals, and may improve its performance with learning or acquiring knowledge

What is a social robot?

A social robot is an artificial agent which has a physical body with some human characteristics (e.g. head, eyes, mouth, arms) that allow it to interact with humans in similar ways that humans communicate and interact with the world (e.g. by talking or using social cues, such as facial expressions or gestures).

What will the project entail?

You will interact with an artificial agent and/or a social robot in a scenario with simulation of an autonomous vehicle, where you will be in the role of the driver of the vehicle. The study will take place at the Uppsala Social Robotics Lab, Department of Information Technology, Uppsala University, and will take approximately 30 minutes of your time.

The study is not intrusive (it will not hurt and you will not be asked to do anything you don't want to). The researchers will record you with video and audio recording. A video camera and a head-worn microphone are used while you interact with the robot.

A copy of these recordings will be used to analyze your behavior while interacting with the robot (for example, how you move and where you look). After the study, you will be asked to fill out questionnaires to share your experience of interacting with the robot during the activity.

Informed Consent form

(Swedish form provided with English translation below)

Samtycke till att delta i projektet
Consent to participate in the project

Jag har fått muntlig och/eller skriftlig information om studien och har haft möjlighet att ställa frågor. Jag får behålla den skriftliga informationen.

- Jag samtycker till att delta i projektet "Etisk och pålitlig interaktion mellan människa-robot och människa-agent i sociala sammanhang med autonoma fordon och drönare"
- Jag samtycker till att uppgifter om mig behandlas enligt det sätt som beskrivs i forskningspersonsinformationen
- Jag godkänner / jag godkänner inte att mina bilder och videoinspelningar kan visas i vetenskapliga artiklar, konferenser och evenemang

I have received oral and/or written information about the study and have had the opportunity to ask questions. I get to keep the written information.

- I agree to participate in the project "Ethical and trustworthy interaction between human-robot and human-agent in social contexts with autonomous vehicles and drones"
- I agree that data about me is processed according to the way described in the research person information
- I agree / I do not agree that my photos and video recordings can be shown in scientific articles, conferences and events

Plats och datum (Place and date)	Underskrift (Signature)
	Namnförtydligande (Name)

2.3.3 Bias, System Auditability, and User Agency

In SymAware's WP6, UU will investigate the requirements for trustworthy and ethical situational awareness in human-agent interaction. The development of socially aware artificial agents and robots must support inclusion and diversity, for example gender diversity, in the entire design cycle of a robot system.

For example, if some of our tasks should require the building a machine learning functionality for a social robot using datasets of people's faces for training purposes, attention to gender, ethnicity, and age diversity will be necessary in order to avoid biases.

In order to tackle this issue and promote fairness, we will aim to ensure the collection of training data from a balanced dataset. We will research best practices in relevant state-of-art literature and carefully take measures to prevent, avoid and mitigate any potential bias, discrimination and stigmatization, both in the input data as well as in the algorithm design.

More specifically, in WP6, UU is coordinating a dedicated task (T6.3) to investigate in-depth how to avoid biases and promote diversity and fairness, at all different stages of the design, development and evaluation of the situational awareness technologies.

Moreover, during the entire lifecycle of the project (see T6.1, T3.2, T4.3), UU will be carrying out participatory design studies with the end users of these technologies. We will investigate if and how the AI-related developments can allow for the functionality to enable external human agencies to audit the communications, decision processes and other aspects of the AI interactions and behavior. We will research and workshop together with the end users on how humans can best maintain meaningful control over the most important aspects of decision-making process involved in the specific use case and specific agent/robot.

These studies will be grounded in the key concepts of Accountability, Responsibility and Transparency [11]:

- ***Accountability*** is the agent's need to explain and justify its decisions and actions to the other agents with whom it interacts. To ensure accountability, decisions must be derivable from, and explained by, the decision-making algorithms used. This includes, for example, the need for representation of the moral values and societal norms relevant in the context of operation, which the agent uses for deliberation. Accountability in AI requires both functionality for guiding action (by forming beliefs and making decisions), and for explanation (by placing decisions in a broader context and by classifying them along moral values).

- ***Responsibility*** refers both to the capability of an agent, and to the role of the users interacting with it. Both need to be considered when accounting for a decision and when diagnosing errors or unexpected results. As the chain of responsibility grows, one important question is how to link the agent/robot's decisions to the fair use of data and to the actions of stakeholders involved in the system's decision.

- ***Transparency*** is the need to describe, inspect and reproduce the mechanisms through which an agent makes decisions and learns to adapt to its environment. Transparency also covers the governance of the data used and created. To ensure the AI algorithms developed and used will not be essentially black boxes, we will investigate how to provide explanation and clarity to the users by relying on methods such as Explainable AI to inspect algorithms and their results and to manage data provenance and dynamics.

3. Related SymAware Deliverables in WP9

3.1 Deliverable 9.2

For every planned study with human participants by UU, a detailed risk assessment will be conducted in collaboration with the independent ethics advisor.

Details on the possible identified ethical concern, strategies to mitigate them, and measures to prevent any potential misuse of the research findings will be submitted as a report before the start of the relevant activities.

3.2 Deliverable 9.3

On March 16th 2023 Matthijs Smakman was appointed in the role of an independent Ethics Advisor in the SymAware project, and will collaborate with all partners and advise the consortium on how to address the ethics issues identified during the project's activities.
Ethics reports with details on how the ethics issues were addressed must be submitted with the periodic reports.
Matthijs Smakman is an Associate Professor at Utrecht University of Applied Sciences and a Lecturer at Vrije Universiteit Amsterdam. His CV is attached in D9.3.

Bibliography

[1] Gray, H. M., Gray, K., & Wegner, D. M. (2007). Dimensions of mind perception. science, 315(5812), 619-619.

[2] Fink, J. (2012). Anthropomorphism and human likeness in the design of robots and human-robot interaction. In Social Robotics: 4th International Conference, ICSR 2012, Chengdu, China, October 29-31, 2012. Proceedings 4 (pp. 199-208). Springer Berlin Heidelberg.

[3] Wiese, E., Metta, G., & Wykowska, A. (2017). Robots as intentional agents: using neuroscientific methods to make robots appear more social. Frontiers in psychology, 8, 1663.

[4] Wallkötter, S., Tulli, S., Castellano, G., Paiva, A., & Chetouani, M. (2021). Explainable embodied agents through social cues: a review. ACM Transactions on Human-Robot Interaction (THRI), 10(3), 1-24.

[5] Panchea, A. M., & Ferland, F. (2020). From Humans and Back: a Survey on Using Machine Learning to both Socially Perceive Humans and Explain to Them Robot Behaviours. Current Robotics Reports, 1, 49-58.

[6] Mitchell, D., Choi, H., & Haney, J. M. (2020, December). Safety Perception and Behaviors during Human-Robot Interaction in Virtual Environments. In Proceedings of the Human Factors and Ergonomics Society Annual Meeting (Vol. 64, No. 1, pp. 2087-2091). Sage CA: Los Angeles, CA: SAGE Publications.

[7] Khavas, Z. R., Ahmadzadeh, S. R., & Robinette, P. (2020). Modeling trust in human-robot interaction: A survey. In Social Robotics: 12th International Conference, ICSR 2020, Golden, CO, USA, November 14–18, 2020, Proceedings 12 (pp. 529-541). Springer International Publishing.

[8] Gao, Y., Sibirtseva, E., Castellano, G., & Kragic, D. (2019, November). Fast adaptation with meta-reinforcement learning for trust modelling in human-robot interaction. In 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 305-312). IEEE.

[9] Winkle, K., Senft, E., & Lemaignan, S. (2021). LEADOR: A method for end-to-end participatory design of autonomous social robots. Frontiers in Robotics and AI, 8, 704119.

[10] Koo, J., Kwac, J., Ju, W., Steinert, M., Leifer, L., & Nass, C. (2015). Why did my car just do that? Explaining semi-autonomous driving actions to improve driver understanding, trust, and performance. International Journal on Interactive Design and Manufacturing (IJIDeM), 9, 269-275.

[11] Dignum, V., Baldoni, M., Baroglio, C., Caon, M., Chatila, R., Dennis, L., ... & de Wildt, T. (2018, December). Ethics by design: Necessity or curse? In Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society (pp. 60-66).